Practical 3: Reading Datasets (Text, CSV, JSON, XML)

Practical Information

Attribute	Details
Practical Number	3
Title	Reading Datasets (Text, CSV, JSON, XML)
Duration	2 Hours
Course Outcomes (COs)	CO2
Description	Read and parse datasets in multiple formats and prepare them for analysis.

Key Topics Covered

- CSV
- JSON
- XML parsing
- File I/O
- Data validation
- Format conversion

Instructions for Students

1. **Complete all tasks** outlined in this practical
2. **Write your code** in the cells provided below
3. **Add comments** to explain your code
4. **Include proper output** (prints, visualizations, etc.)
5. **Submit your completed notebook** before the deadline

Your Work Starts Here

Use the cell(s) below to write your implementation:

```
# Practical 3: Reading Datasets (Text, CSV, JSON, XML)
# Write your implementation here
import pandas as pd
import numpy as np
import json
import xml.etree.ElementTree as ET
from sqlalchemy import create_engine, inspect
```

```
import sqlite3
import time
import os
import sys
from pathlib import Path

# Print versions
print("📚 LIBRARY VERSIONS:")
print(f"Python: {sys.version.split()[0]}")
print(f"Pandas: {pd.__version__}")
print(f"NumPy: {np.__version__}")
print("\n✅ All libraries imported successfully!")

# Set up display options
pd.set_option('display.max_columns', None)
pd.set_option('display.max_colwidth', None)
pd.set_option('display.width', None)
```

```
📚 LIBRARY VERSIONS:
Python: 3.13.5
Pandas: 2.2.3
NumPy: 2.1.3

✅ All libraries imported successfully!
```

```
# Sample CSV data for demonstration
sample_csv_data = '''ID,Name,Age,Department,Salary
1,Shravya,25,Engineering,75000
2,Bhavya,29,Sales,65000
3,Adda,29,Marketing,60000
4,Roshani,30,Engineering,80000
5,Priya,26,HR,55000'''

# Write to CSV file
with open('sample_data.csv', 'w') as f:
    f.write(sample_csv_data)

print("✅ Sample CSV file created!")
print("\nFirst few lines of CSV:")
print(sample_csv_data[:200] + "...")
```

```
✅ Sample CSV file created!

First few lines of CSV:
ID,Name,Age,Department,Salary
1,Shravya,25,Engineering,75000
2,Bhavya,29,Sales,65000
3,Adda,29,Marketing,60000
4,Roshani,30,Engineering,80000
5,Priya,26,HR,55000...
```

```
# Sample JSON data
sample_json_data = [
    {"ID": 1, "Name": "Shravya", "Age": 25, "Department": "Engineering", "Salar
    {"ID": 2, "Name": "Bhavya", "Age": 29, "Department": "Sales", "Salary": 65(
    {"ID": 3, "Name": "Adda", "Age": 29, "Department": "Marketing", "Salary": 6
    {"ID": 4, "Name": "Roshani", "Age": 30, "Department": "Engineering", "Salar
    {"ID": 5, "Name": "Priya", "Age": 26, "Department": "HR", "Salary": 55000}
]

# Write to JSON file
with open('sample_data.json', 'w') as f:
    json.dump(sample_json_data, f, indent=2)

print("✅ Sample JSON file created!")
print(f"\nJSON content preview:")
print(json.dumps(sample_json_data[:2], indent=2))
```

```
✅ Sample JSON file created!

JSON content preview:
[
  {
    "ID": 1,
    "Name": "Shravya",
    "Age": 25,
    "Department": "Engineering",
    "Salary": 68000
  },
  {
    "ID": 2,
    "Name": "Bhavya",
    "Age": 29,
    "Department": "Sales",
    "Salary": 65000
  }
]
```